Problem 4 – Bill of Materials for an RC Aircraft

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Aircraft brief and assumptions

Electric, single-motor, fixed-wing RC trainer: about 1.1 m wingspan, high wing, hand launch, lightweight foam structure. No parts were purchased for this assignment.

- Prices are in INR including GST. Items marked * are read from a listing (see Sources). All other prices are my market estimates and must be replaced with verified links before submission.
- Mean wing chord 0.2 m, so wing area is about 22 dm².
- Foam sheet mass is assumed at about 55 g (Depron 6 mm) and 85 g (foam board 5 mm) per sheet.
- Static thrust (800 g budget, 900 g performance) and full-throttle current (14 A and 15 A) are assumptions, not measurements. They must be verified on a thrust stand.
- A 10% mass contingency covers glue, tape and cable slack.

Part A – Identify components

Airframe and reinforcement(reinforcement helps to make plane more stronger and by adding structural support)

- Foam sheets (6 mm Depron/XPS) for fuselage, wing and tail, about 4 sheets
- 6 mm carbon-fibre tube spars (wing spar and fuselage keel); wooden dowel as the cheaper alternative
- Hinge and covering tape

Propulsion(Battery → ESC (CONTROLS MOTOR SPEED) →Electric Motor +Propeller (converts electrical energy into thrust) → Thrust)

- Brushless motor, 2212 class
- ESC, 30 A, with 5 V BEC
- Propeller (9×4.7 or 10×4.5 slow-fly) plus spares
- Prop adapter and motor mount with M3 screws

Power

- 3S 2200 mAh LiPo battery
- Balance charger (ground equipment, not airborne)
- Battery strap and LiPo voltage alarm

Control

- FlySky FS-i6 transmitter with FS-iA6B receiver (2.4 GHz)
- 4 servos: 2 ailerons, elevator, rudder(servo help us to control the direction where plane goes)
- Control horns, pushrods and clevises
- Servo extension and Y leads

Landing gear

- None for the chosen build (hand launch, belly landing). A wire gear with two wheels is an option for ground launch.

Connectors, wiring and assembly materials

- XT60 connectors (battery to ESC) and 3.5 mm bullet connectors (ESC to motor)
- Heat-shrink, zip ties, spare wire
- Foam-safe adhesives (hot-melt glue or UHU Por). Ordinary CA or solvent glue dissolves Depron.
- And multimeter to check the connections and voltage .

Part B – Compare options

Component	Recommendation 1	Recommendation 2	Key specifications	Qty	Estimated cost (₹) Rec 1 / Rec 2
Motor	A2212 1000KV*	Readytosky 2212 920KV*	2212 class, 6 mm shaft, 7–12 V rated, 30 A ESC, 9×4.7 or 10×4.5 prop, 55–60 g	1	371 / 660
ESC	SimonK 30A*	Branded 30A with switching BEC	30 A continuous, 2–4S, 5 V BEC	1	342 / 1,500
Battery	3S 2200 mAh 25C	3S 2200 mAh 35C	11.1 V, 55 A / 77 A capability, XT60, about 180 g	1	1,300 / 1,900
Propeller	9×4.7 slow-fly	10×4.5 slow-fly	Matches motor KV on 3S; includes 2 spares	3	270 / 480
Transmitter + receiver	FlySky FS-i6 + iA6B*	FlySky FS-i6X + iA6B*	2.4 GHz AFHDS 2A, 6 ch / 10 ch	1	4,056 / 6,210
Servos	SG90 (plastic gear)	MG90S (metal gear)	9 g class, 4.8–6 V	4	440 / 1,000
Airframe foam	Depron/XPS 6 mm	Foam board 5 mm	About 55 g vs 85 g per sheet	4	880 / 520
Spars	6 mm carbon tube	6 mm wooden dowel	Wing spar and fuselage keel	2	700 / 80
Other components	Pushrods, mount, adapter, XT60 and bullet connectors,	Better-grade versions of the same parts	Foam-safe glue only; charger is ground equipment	set	2,610 / 6,800

Component	Recommendation 1	Recommendation 2	Key specifications	Qty	Estimated cost (₹) Rec 1 / Rec 2
	leads, heat-shrink, glue, tape, strap, LiPo alarm, charger				
Total (all Rec 1 / all Rec 2)					**10,969 / 19,150**

Part C – Justify selections

1. Motor–propeller match and thrust check. A 1000 KV motor on 3S (11.1 V) spins at about 11,000 rpm unloaded and about 8,800 rpm loaded (75–80%). A 9×4.7 prop then has a pitch speed of about 8,800 × 4.7 × 0.0254 / 60 ≈ 17–18 m/s. That is well above the roughly 9 m/s stall speed of a 1.1 m trainer and allows cruise at about 12–15 m/s. A larger, lower-pitch 10×4.5 prop gives more static thrust and suits the 920 KV option.

to check the available thrust : we have two methods (a)Physical test by digital scale or fish scale method : in this we mount a digital luggage scale to a heavy table or wall and connect a non stretching wire from the scale to the tail or a secure mounting point near the rear of the rc plane make sure the plane can slide forward slightly and then power up the transmitter to full throttle and read the weight /force (in grams) :

B Estimation :

By using RC Planes online static thrust calculator we have to just input our propeller diameter , pitch , and estimated RPM. and it will give us the percentage by sometimes it calculates wrong performances 15 to 30 percent due to battery and ESC limits and air turbulence so physical testing is always preffered.

2. ESC rating and electrical margin. Expected full-throttle current is about 14–15 A (assumed). With the rule that the ESC should be rated at least 1.5–2 × the maximum expected current, a 30 A ESC gives about 2× margin, covering voltage sag and prop changes. It must support 3S, and its BEC must power four micro servos and the receiver at 5 V. The cheap SimonK listing does not state a BEC rating, so the branded ESC is preferred for the competition build.

3. Battery selection and trade-offs. 3S matches the motor's 7–12 V rating in the vendor listing. 4S would exceed that rating and need a smaller prop; 2S is underpowered. Each extra 100 mAh adds roughly 8 g, which raises endurance but lowers thrust-to-weight and raises wing loading. At 25C × 2.2 Ah the pack can supply 55 A against a demand of about 14 A, so a higher C rating mainly adds cost and mass. The 35C pack is chosen for lower voltage sag and reliability, not necessity. The pack is positioned with the strap so the CG falls at 25–33% of the chord.

4. Servo count and control surfaces. Four servos: two ailerons (on a Y-lead), one elevator and one rudder. Throttle runs through the ESC on its own channel. A rudder-elevator-throttle layout would save two servos (about ₹220–500 and 18–27 g) but reduces roll authority, so ailerons are kept.

5. Structural material.

- Depron/XPS 6 mm: very light, cheap, repairable in minutes with foam-safe glue, but flexible, so a carbon spar and tape covering are added. Depron's dense closed-cell structure provides great structural strength.
- Foam board 5 mm: cheaper and stiffer but about 55% heavier, which cuts thrust-to-weight.
- Balsa: light and stiff for its weight, but fragile, costlier and slower to repair.

Selected: Depron 6 mm skins, 6 mm carbon spar, foam-safe glue and tape.

6. Electrical and mechanical compatibility checks.

- *Electrical:* pack voltage (3S, up to 12.6 V) within motor and ESC ratings (the motor listing states 7–12 V, so confirm against the datasheet); ESC continuous current at least 1.5 × motor current; battery C-capability above demand; matching XT60 connectors and correct polarity; 5 V BEC within the servo (4.8–6 V) and receiver (4.0–8.4 V) ranges.
- *Mechanical:* motor shaft diameter against prop adapter; motor mount hole pattern; prop clearance from fuselage and ground; servo bay fit and pushrod travel; CG at 25–33% of chord.
- *Transmitter–receiver:* the FS-iA6B is listed as compatible with the FS-i6 (same AFHDS 2A protocol, 2.4 GHz). Bind them, set failsafe to idle throttle, check all channel directions with the prop off, and do a walk-away range test before the first flight.

Part D – Budget

1. Total build cost.

Configuration	Total cost (₹)
Budget-oriented	10,349
Performance-oriented	15,176

Both totals include the balance charger. If every Rec 1 is chosen the total is ₹10,969, and if every Rec 2 is chosen it is ₹19,150.

2. Safe cost-saving opportunities.

- Depron plus wooden dowel spars for non-critical structure.
- Generic A2212 motor with a tested ESC.
- SG90 servos on the budget build.
- Reuse a club charger or transmitter.
- Hot-melt glue, locally bought foam, and bulk spare props.
- Do not save on LiPo quality, ESC BEC quality, connectors and soldering, receiver failsafe, or foam-incompatible glue.

3. Budget-oriented and performance-oriented configurations.

Item	Budget	Performance
Motor and prop	A2212 1000KV, 9×4.7	2212 920KV, 10×4.5
ESC	SimonK 30A	Branded 30A with known BEC
Battery	3S 2200 mAh 25C	3S 2200 mAh 35C
Servos	SG90	MG90S metal gear
Spars	Wooden dowel	Carbon tube
Connectors and consumables	Standard	Better grade

4. Choice for the competition: performance-oriented. It costs about ₹4,800 more (roughly 45%). In return it removes the likeliest in-competition failures: BEC brown-out with an unspecified ESC, stripped plastic servo gears, and voltage sag from a weak pack. If a hard budget cap applies, the budget build is the fallback; the same airframe works for both.

Bonus – Takeoff mass and propulsion sufficiency

	Budget	Performance
Airborne mass (before contingency)	700 g	742 g
Takeoff mass (+10% contingency)	770 g	816 g
Wing loading (22 dm² wing)	35 g/dm²	37 g/dm²
Assumed static thrust	800 g	900 g
Thrust-to-weight ratio	1.04	1.10
ESC margin (30 A ÷ full-throttle current)	2.1×	2.0×
Estimated endurance	about 15 min	about 14 min

Assessment. On the stated assumptions, propulsion is sufficient: thrust-to-weight is just above the 1.0 target for both builds, and wing loading is within the usual trainer range of about 20–40 g/dm². This is not guaranteed. The vendor listing for the 920 KV motor quotes only about 500 g of thrust, and at that figure thrust-to-weight falls to about 0.6, too low for a safe hand launch. The motor and prop must therefore be tested on a thrust stand before committing to the build.